

Homework Assignment #1

Due: Tuesday, September 22, 2026, 5:00pm (EST)

- Homework assignments can be done in groups of three or fewer people. Only turn in one write-up per group.
- In your write-up, please:
 - Include (preferably on the first page) a summary of your answers to all the questions.
 - Give your answers to at least five digits (if appropriate) after the decimal place (this makes it easier to spot which answers are correct or where mistakes occur).
- In your solutions to the individual questions, please:
 - Explain the main steps or formulas used to arrive at your solution. Specifically, *show* and *explain* the main formulas used in any spreadsheet or software output you hand in. The formulas in the annotated output can be handwritten (do not have to be typeset).
 - Include row and column headings in spreadsheet printouts (in Excel, choose *File—Print Preview*, then click on *Setup*, then choose the *Sheet* tab and check the “row and column headings” box).
- Please organize and structure software output you hand in so that it is easy for others to understand. Spend extra time, after getting the answers correct, to create clean-looking, understandable submissions.
- It is not sufficient to turn in a spreadsheet or some other software output (in printout or electronic form) with only numbers showing. Points will be deducted for write-ups turned in without the main formulas shown and explained, even if the numerical solutions are correct. How much detail is needed in the explanations? Enough detail so that anyone else in the course (student, TA, instructor) could follow (and, relatively easily, reproduce) your work and final solutions using only your write-up.
- Late homework submissions will not be accepted.

1. (*AI Tools*) In this assignment, you use one of the multiple AI tools commonly available today to create and test a systematic investment strategy from scratch.

A recent paper by Zarattini, Aziz, and Barbon, *Beat the Market: An Effective Intraday Momentum Strategy for S&P500 ETF (SPY)* (available for download from the link in the assignment prompt and from https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4824172), describes a systematic intraday trading strategy on the SPY. In the context of this strategy, the paper uses some concepts we will see in the course of the semester: potential improvements to the original version of a strategy (one through a stop/loss rule, another through a volatility adjustment for leverage control), the notion of a dynamic daily range within which a stock may be acting differently, etc. Yet, since we have not covered this particular strategy (or any other strategy of similar nature) in class, replicating this strategy is akin to developing an idea from scratch.

To help you in the development of the strategy, you will use one of the AI platforms widely available. You can choose any platform of your choice, whether one of the ‘first generation’ (bad term!) platforms like ChatGPT, Gemini, Copilot, Perplexity, Claude, Meta AI, Grok, etc. or one of the ‘second generation’ (again, bad term!) Vibe Coding platforms like Cursor, Windsurf, Lovable, Replit, Bolt, v0, Base 44, etc. The only requirement for the platform you choose is that you document your choice (and the reasons for it) in your write-up.

You will test your AI implementation of the strategy on some actual data (see below) and report your findings. You will also critique the strategy (as an idea), its implementation (how good a job the AI tool you have chosen has done), and, if possible, suggest and demonstrate potential improvements (to both the strategy itself and its AI implementation by the tool you have selected).

Step 1

You are to upload the paper into one of the available AI tools and coax the AI tool to produce an implementation that replicates the strategy of the paper. For this step:

- a) You can use any AI tool of your choice but you must use an AI tool to produce the code the implements the strategy.
- b) You can choose the language (or software environment) for the output that the AI tool produces (e.g., Python, Matlab, R, Excel, any other, possibly more advanced, software environment of your choice).
- c) You cannot use code that you write or the code that the authors provide. You can use these alternatives only to check the “goodness” of the code that the AI produces and/or to help correct, or improve, the AI-produced replicator.
- d) You must check the accuracy of the final version of the AI-produced replication of the strategy: you must verify that the code produces correct results. For this check, you may need a version of the code you create, the code the authors provide, or similar code produced by another AI tool. You can also do spot-checking by doing “localized” (small scale) manual calculations.
- e) You can choose which parts of the paper you coax the AI tool to reproduce. The paper contains various possible threads, in terms of tests, improvements, and variations of the strategy. You can select which ones you replicate and which ones to ignore, but you have to explain the reasons for your selection.

Step 2

You are to test your AI-generated replication of the strategy in the paper with some real data. The file `hw1.spy.20250313-20251201.intra-30m.xlsx` contains approximately 8 months of intra-day trading data for SPY, the financial instrument used in the strategy of the paper. The data, for the period March 2025 – November 2025, consist of the ending prices and the accumulated volumes of SPY at 30' intervals. This sampling frequency of 30' intervals has been chosen to match the frequency used in the paper.

The data have been collected from the Bloomberg terminal, and have been cleansed and organized through what, unfortunately, proved to be a very arduous and painful process that required multiple iterations. There may still be problems even with the cleansed data; please identify and report any data anomalies you happen to identify or you may think you identify. And there are two 'special' days in the dataset: 7/3/2025 and 11/28/2025; on these two days the US stock market closed at 1pm (because of the holidays on the following and preceding days respectively).

The data provided amount to the maximum amount of intra-day data one can collect as part of the standard Bloomberg terminal service. These data may not be sufficient for a thorough testing of the strategy (e.g. the daily patterns the paper explores in relation to the strategy). You are welcome to use any other relevant dataset of your choice, perhaps of greater accuracy and/or covering a longer time period, as long as you specify which data you have used (and their source) and you submit your data with your homework.

Step 3

For your homework assignment, please deliver:

1. A clear and detailed description of your Step 1.

In your writeup, please provide a detailed account of the process you have followed to replicate the strategy: which AI tool you have selected (and why), which parts of the paper you have chosen to replicate and which not (and why), what prompts you have used, what iterations you have had to go through in order to arrive at the final product (and why), how accurate you found the AI-generated replication to be, what corrections it required and how you effected those iterations, etc.

2. A detailed analysis of your findings in Step 2.

In your writeup, please describe the data you have used to evaluate the strategy and the results of your replication of the strategy on those data. The analysis should include all the return statistics you consider relevant, any comparable statistics (if any), and a robustness discussion (vis-à-vis any choices, parameters, or other dependencies you consider significant). The analysis should also include a discussion of any relationship, in either direction, between the data you used and the parts of the paper you chose to replicate.

3. An assessment of the strategy itself.

What do you think of the main idea, the variants, and the improvements the paper suggests? Is the strategy interesting and/or attractive? If so, for what kind of investors? Is this a strategy you could implement (and use) on your laptop? What conditions, you think, are critically relevant to the (attractive) properties of the strategy? Can you identify market conditions under which you would expect the strategy to be more, or less,

profitable? How does the strategy relate to strategies or concepts we have discussed in class or have been presented in the papers we read? If you were to apply the same strategy to the index of another major market, would you expect the returns of the two applications to be similar or different? Are there any glaring risks to the strategy that are not presented in the paper? Have you thought of any improvements for the strategy?

4. A critical review of the paper.

Did you find the paper interesting? Useful? Realistic? What did you think of the paper's analysis of return patterns? Would you have written the paper (or presented its results) differently? If so, how?

Note

The homework is intended to be an open-ended experiment through which you will get some experience in using the ubiquitous AI tools to craft and explore an investment strategy. Your submission will be judged more by its thoroughness and completeness than by your strategy's success (or not) with the data you use to test it.